

A STEPWISE METHOD FOR THE IDENTIFICATION OF MULTIPLE OUTLIERS AND INFLUENTIAL OBSERVATIONS

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Key words: Algorithm; leverage points; multiple-case influence measures; number of influential observations; regression.

Summary: A stepwise method is proposed to reduce the large computational burden for searching for multiple outliers and influential observations in regression. The method has an intuitive appeal for applied statisticians and can be employed easily. Its performance in analysing thirty-five data sets, of various sizes and dimensions, which commonly appear in the literature of regression diagnostics and robust regression, is promising. The method can successfully detect four high leverage points in a data set of which a high-breakdown robust estimator has difficulties. The saving in computational effort relative to the usual complete enumeration approach when searching for a group of up to four outliers and influential observations in a pollution data set of size 117 was of the order of one hundred-fold.

1. INTRODUCTION

The identification of outliers and influential observations in multiple regression is important and is regarded as a useful procedure in model criticism (Box, 1980). The research in this field has been extensive in the last decade. The residual and the leverage statistic are two measures commonly used in the identification process. Indeed, more than twenty measures with different rationale and reasoning have been proposed and they can be found in the four survey books of the field, namely Belsley et al. (1980), Cook and Weisberg (1982), Atkinson (1985), and Chatterjee and Hadi (1988). However, the literature is mainly about single-case influence measures which are evaluated through the omission of individual observa-

tions. It is well known that applying the single-case measure sequentially may not be able to indicate multiple outliers and influential observations because of the masking effect problem. This problem has been addressed by Gray and Ling (1984) using the cluster analysis approach, Atkinson (1986) and Rousseeuw and van Zomeren (1990) using the robust approach, and Cook (1986) using the local influence approach. Rousseeuw and Leroy (1987) in their book suggested using the high-breakdown robust methods. Searching for a block of multiple outliers and influential observations using the multiple-case influence measure is a natural extension of the single omission approach and is often referred to. But this block search procedure involves a very heavy computational load for moderate and large sized data sets, so practically it is difficult to employ.

In this paper, we consider specifically the computational problem of the block search approach. A stepwise method which can greatly reduce the heavy computational burden is proposed in section 2. It is easy to use and has an intuitive appeal for applied statisticians. It is applied to search for the most influential sets of all the fourteen relevant data sets from the four regression diagnostic books previously described. A good performance is obtained (section 3). Twenty data sets (Rousseeuw and Leroy, 1987) generally having more structure are analysed by the proposed method. Satisfactory results are also obtained. The four high leverage points in the modified wood specific gravity data set, which cannot be determined using a high-breakdown method, are successfully detected by the stepwise method (section 4). A data set of size 117 is analysed by this method in section 5 with satisfactory results and with a large saving in computational effort.

2. MULTIPLE-CASE INFLUENCE MEASURES AND THE ALGORITHM

A multiple regression model is written as $y = X\beta + \epsilon$ where y is an $n \times 1$ vector of values of dependent variable, $X = (x_1, x_2, \dots, x_n)^T$ is an $n \times p$ full rank matrix of fixed and known independent variables, β is a p -vector of parameters, and ϵ is an $n \times 1$ vector of errors distributed as $N(0, \sigma^2 I)$. The usual maximum likelihood estimator for β is $\hat{\beta} = (X^T X)^{-1} X^T y$ and the

fitted values are $\hat{y} = X \hat{\beta}$. An unbiased estimator for σ^2 is given as $s^2 = (y - \hat{y})^T (y - \hat{y}) / (n - p)$ where $y - \hat{y} = e$ is the vector of error residuals. This vector has zero mean and variance, $\sigma^2(I - H)$, where $H = X(X^T X)^{-1} X^T$ is normally called the hat matrix with the diagonal element h_{ii} being the leverage measure (Hoaglin and Welsch, 1978).

There are more than twenty different influence measures proposed for detecting outliers and influential observations, but only a few of the more representative ones are selected in this study. Basically, the measures we are going to study are selected from two broad types (Chatterjee and Hadi, 1988, chapter 5). One refers to the change of the volume of confidence ellipsoids, for which the form can be written as a linear function of $U_I = V_{(I)}/V$, where V and $V_{(I)}$ are certain statistics based respectively on the full data set and the data set with an index set I of size m omitted. The generalized leverage GL_I (Hoaglin and Welsch, 1978), the outlier statistic OUT_I (Barnett and Lewis, 1984), the Andrews-Pregibon statistic AP_I (Andrews and Pregibon, 1978) and the covariance ratio $COVRATIO_I$ (Belsley et al., 1980) are selected. Their forms are given in table 1. It should be noted that, unlike GL_I , OUT_I and AP_I where only large values of the statistics indicate influential points, large as well as small $COVRATIO_I$ values would indicate that observations with indices I are influential. Another type of measure which corresponds to the change in regression coefficients may be written as (Cook and Weisberg, 1982, p. 114), $D_I(L, G) = (\hat{\beta} - \hat{\beta}_{(I)})^T L (\hat{\beta} - \hat{\beta}_{(I)}) / G$, with different choices of L and G . The generalized Cook's C_I (Cook, 1977), the generalized Welsch-Kuh's WK_I (Welsch & Kuh, 1977) and the generalized Welsch's W_I statistics are of this type (see table 1 for the forms). GL_I and OUT_I are used for detecting high leverage and outlying observations respectively. Whereas, the other measures are useful for the detection of influential observations which may be of high leverage and/or outlying.

Consider an influence measure T_J . Let J^* be the subset of size $m + 1$ that gives the largest value of T_J (or the smallest for $COVRATIO_J$). That is, $T_{J^*} = \max_J(T_J)$. Let $I^\#$, where $I^\# \in J^*$, be the subset of size m ,

with $T_{I\#}$ being the largest among the set $\{T_I, I \in J^*, \text{ size of } I = m\}$ which contains $m + 1$ elements. From practical experiences, it is often found that $T_{I\#}$ is the largest or one of the largest among all possible T_I (see the many examples in sections 3, 4 and 5). Thus, it seems reasonable to search for $T_{J*} = \text{maximum}(T_J)$ based on the M index sets (the choice of M is discussed later) giving largest T_I values. The following stepwise algorithm is suggested for searching for multiple outliers and influential observations:

- (a) Start with the size of influential set, $m = 1$. Find the M most influential observations.
- (b) Add each of the $n - m$ observations to each of the M most influential sets of size m . Evaluate the measure values.
- (c) Find the M most influential sets in (b). Stop if the predetermined maximum size of influential sets is reached; increase the size m by one and go to (b) otherwise.

Note that duplication could occur in (b) as in the following illustration: If the sets $\{1, 2\}$ and $\{1, 3\}$ are both members of the M influential sets, the set $\{1, 2, 3\}$ would be evaluated twice. This duplication has to be avoided for reducing the computational load.

Some analytical justifications of the algorithm are given in the appendix. We know, from the discussion there, that the algorithm would have a better performance for the confidence ellipsoid measures. The main justification of the algorithm, however, lies in its good performance in searching for outliers and influential observations in various data sets to be discussed later. Fung (1988) implicitly used the method for testing multivariate outliers. The value of M for the algorithm needs to be chosen, but the "optimal" value of M depends on n, m, p , the data structure and the specific form of the measure. Since the algorithm would perform better for the change of confidence ellipsoid measures, so M could be smaller for these measures. Moreover, the relative size of M to n could be smaller for larger data sets

as in this case the measures are more stable. This implies that the number of searches would still be of acceptable size even when n is very large. Of course M should be larger for larger m and p . From practical experiences, we find the following choice suitable. Take M to be $100m$ or $200m$ if n is less than a hundred, and $200m$ or $400m$ otherwise. However, when n is small, say smaller than 30, the computational cost for a complete enumeration is indeed minimal. Note that the computational cost for the algorithm, in searching for multiple influential observations, is only a linear function compared to the exponential function of n for the complete enumeration approach.

3. PERFORMANCE OF THE STEPWISE METHOD FOR FOURTEEN DATA SETS

The performance of the algorithm is investigated using all the fourteen data sets (most are real) that have been analysed for outliers and influential observations in the four surveys books on regression diagnostics. The eight measures given in table 1 are evaluated for these data sets with a complete enumeration for $m = 1, \dots, 4$. Let I_m and I_{m+1} , having respectively m and $m + 1$ elements, be the most influential sets for a particular measure. One important finding is that I_m is usually a subset of I_{m+1} for $m = 2$ and 3 (as mentioned in section 2). There are 47 out of 224 cases ($=14 \times 8 \times 2$) not belonging to this. For each of these cases, we put the smallest ranking that is required for searching for the most influential set of the next order (see table 2). The rankings are generally low. (This implies that the ranking for $T_{I^\#}$ is low as discussed in section 2.) From the table, it is observed that a larger M is required for a larger m as expected. Moreover, measures C_I , WK_I and W_I need a larger M as previously discussed. The stepwise algorithm does perform well in finding all the most influential sets (up to $m = 4$) in all fourteen data sets for the measures. Taking $M = 100$ is actually sufficient for these data sets. It is to be noted that the algorithm can also find the influential set (3,20) which, when it is omitted, gives an exact multicol-

linearity or a rank deficiency for the explanatory variables X in CW4. Moreover, the removal of observations 5, 6, 9 and 26 would also lead to a near or exact multicollinearity for X in CW149.

4. ROUSSEEUW AND LEROY'S DATA SETS

It has been suggested that the algorithm should be tried on the Rousseeuw and Leroy's (1987) data sets which generally have more structure. Twenty relevant data sets are analysed up to four outliers and influential observations using the complete enumeration approach and the stepwise algorithm. The two methods give exactly the same results, but the stepwise method could largely reduce the computational load. For brevity, the results are not shown here.

Cook and Hawkins (1990) showed that the commonly used random search procedure for the high-breakdown minimum volume ellipsoid method is not efficient when searching for the four high leverage influential points in the modified wood specific gravity data set (Rousseeuw and Leroy, 1987, p. 243). The procedure requires a large number of evaluations but gives an unclear pattern of leverage points about the data set. Recently, Atkinson (1993) discusses the issue and suggests an algorithm for determining outliers and leverage observations.

The data set is analysed using the stepwise method. Observations 4, 6, 8 and 19 are found to be high leverage influential observations according to GL_I and C_I . Taking M to be 100 and 200 would be enough for C_I and GL_I respectively. It is to be noted that the proposed method successfully identifies the observations which have been modified by Rousseeuw and Leroy (1987).

5. EXAMPLE ON POLLUTION

This example is a data set of size 117 and with eleven predictor variables obtained from a study by Gibbons et al. (1987) about the effect of air pollution to the mortality rate. These many explanatory variables would create a computational difficulty in evaluating the influence measures by the

complete enumeration approach. We instead do a stepwise regression and find four significant factors, namely SMIN, PMEAN, GE65 and PNOW. More discussions and explanations about the data set are given in Gibbons et al. (1987).

The data set with these four independent variables is illustrated using the stepwise method. We search for up to four influential observations. The CPU time required for obtaining the most influential sets for each measure is about six seconds on average using a VAX 6420 mainframe. The corresponding CPU time using the complete enumeration method is about twelve minutes on average. Thus, more than a hundred-fold reduction in the computational load is achieved using the stepwise algorithm. Of course, the computational effort would be much more reduced by using the algorithm if we search for up to 5% (6) or 10% (12) influential observations. The two methods give exactly the same results here. Four most extreme sets for each measure are given in table 3. One important feature of the table is that, for each measure, the most extreme set of size m is just a subset of the most extreme set of size $m + 1$, $m = 1, 2, 3$. By an ad-hoc comparison of the relative magnitude of the measure for the most extreme set to those for the less extreme sets, we are able to identify the outliers and influential observations. We find that observations 21, 85 and 86 are jointly outlying according to measure OUT_I . It is also noticed that (21, 85, 86) is a subset of each of the 114 most extreme sets of size 4 for OUT_I , and so it gives support to our claim that this set is outlying. This coincides with the findings by Gibbons et al. (1987). From table 3 and some further analyses, we find that observations 21, 85, 86 are influential for $COVRATIO_I$.

6. CONCLUDING REMARKS

Because of the heavy computational load required for the complete enumeration approach, the data sets described above are analysed up to four extreme observations. Thirty-five data sets of various sizes and dimensions, of which many of them commonly appear in the literature of diagnostics and robustness, are analysed. These data sets are mostly real and some of them

are artificially constructed or modified for having more structure. However, the proposed stepwise method can successfully identify the extreme observations. Searching for more than four extreme observations using the algorithm has been tried for some data sets, the results seem satisfactory. Some other sets have been attempted, and the algorithm is able to detect the extreme observations correctly. It is to be noted that the algorithm can also be used for other less common influence measures which are usually functions of or in similar forms to the measures in table 1.

Some analytical justifications have been given, but the main justification of the algorithm lies in its good performance as described above. Moreover, the proposed stepwise method has an intuitive appeal for applied statisticians, and can be implemented in computers without much difficulty. Knowing the most influential sets of various sizes provides useful information about the influence structure of the data set. The method could reduce the computational effort to a very large extent, and so it is recommended, at least in exploratory studies, for studying multiple outliers and influential observations in regression.

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APPENDIX

Some analytical justifications for the stepwise algorithm are given below:

The generalized leverage GL_J and the Cook's statistic C_J , one from each category of the measures, are selected for illustration. Write $GL_J = 1 - R_J$ where $R_J = |X_{(I)}^T X_{(I)}| / |X^T X|$. A small value of R_J may indicate large joint-leverage. Consider

$$(A.1) \quad R_J = R_{I+j} = \frac{|X_{(I)}^T X_{(I)}|}{|X^T X|} \cdot \frac{|X_{(I+j)}^T X_{(I+j)}|}{|X_{(I)}^T X_{(I)}|},$$

where $J = I + j$ means that $m + 1$ observations including the j -th and those of subset I are omitted from the sample. Equation (A.1) can be written as

$$(A.2) \quad R_{I+j} = R_I \left[1 - \mathbf{x}_j^T (X_{(I)}^T X_{(I)})^{-1} \mathbf{x}_j \right]$$

which can further be expanded as,

$$(A.3) \quad R_I \left\{ 1 - \mathbf{x}_j^T (X_{(I-i)}^T X_{(I-i)})^{-1} \mathbf{x}_j - \frac{[\mathbf{x}_j^T (X_{(I-i)}^T X_{(I-i)})^{-1} \mathbf{x}_i]^2}{1 - \mathbf{x}_i^T (X_{(I-i)}^T X_{(I-i)})^{-1} \mathbf{x}_i} \right\},$$

where $i \in I$. Let $I^\#$ and J^* be defined similarly as in section 2, that is $\{J^*: R_{J^*} = \text{Minimum}_J(R_J)\}$, and $\{I^\#: R_{I^\#} = \text{Minimum}_I(R_I), I \in J^* = I^\# + j$.

We are interested in studying the conditions that make $R_{J^*} = R_{I^\# + j}$ small.

They are, as seen from (A.3), large $\mathbf{x}_j^T (X_{(I^\# - i)}^T X_{(I^\# - i)})^{-1} \mathbf{x}_j$, large $\mathbf{x}_i^T (X_{(I^\# - i)}^T X_{(I^\# - i)})^{-1} \mathbf{x}_i$, large $(\mathbf{x}_j^T (X_{(I^\# - i)}^T X_{(I^\# - i)})^{-1} \mathbf{x}_i)^2$, and/or small $R_{I^\#}$. We study them individually.

First, $\mathbf{x}_j^T (X_{(I^\# - i)}^T X_{(I^\# - i)})^{-1} \mathbf{x}_j$ would not be too large, otherwise i would be replaced by j to form the subset $I^\#$ as $R_{I^\#}$ is the smallest among the set $\{R_I, I \in J^*\}$. Second, when $\mathbf{x}_i^T (X_{(I^\# - i)}^T X_{(I^\# - i)})^{-1} \mathbf{x}_i$ is large, it would imply that $R_{I^\#}$ is small (see (A.2)). Third, $(\mathbf{x}_j^T (X_{(I^\# - i)}^T X_{(I^\# - i)})^{-1} \mathbf{x}_i)^2$ should not be large since it is of lower order relatively. Fourth, $R_{I^\#}$ is small. Thus, the condition that makes R_{J^*} to be small is that $R_{I^\#}$ is small. Therefore, the algorithm that suggests searching for R_{J^*} among the index sets giving smallest R_I values seems reasonable.

Next we consider the scaled generalized Cook's statistic,

$$ps^2 C_{I+j} = (\hat{\beta} - \hat{\beta}_{(I+j)})^T X^T X (\hat{\beta} - \hat{\beta}_{(I+j)}),$$

which can be expanded as,

$$(A.4) \quad (\hat{\mathbf{y}}_{(I)} - \hat{\mathbf{y}}_{(I+j)})^T (\hat{\mathbf{y}}_{(I)} - \hat{\mathbf{y}}_{(I+j)}) + (\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(I)})^T (\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(I)}) \\ + 2(\hat{\mathbf{y}}_{(I)} - \hat{\mathbf{y}}_{(I+j)})^T (\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(I)}).$$

The first two terms of (A.4) are non-negative, but the last term could be of either sign. For m greater than one, as $\hat{\mathbf{y}}$ and $\hat{\mathbf{y}}_{(I)}$ differ due to the removal of m observations whereas $\hat{\mathbf{y}}_{(I)}$ and $\hat{\mathbf{y}}_{(I+j)}$ differ because of the removal of one further observation, so the elements of $\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(I)}$, for some subsets I , could generally be of larger magnitude than those of $\hat{\mathbf{y}}_{(I)} - \hat{\mathbf{y}}_{(I+j)}$. Therefore, for these subsets I , especially those giving large $C_I = [(\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(I)})^T (\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(I)}) / ps^2]$, the second term of (A.4) would play an important role in determining C_{I+j} . Thus we also suggest using the algorithm as described in section 2 to search for Max C_I . However, it is expected that the performance of the stepwise method for C_I would normally not be as good as that for GL_I .

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TABLE 1: Selected representative influence measures

Measures	Forms
1. GL_I	$1 - X_{(I)}^T X_{(I)} / X^T X $
2. OUT_I	$1 - s_{(I)}^2 / s^2$
3. AP_I	$1 - Z_{(I)}^T Z_{(I)} / Z^T Z^* $
4. Large $COVRATIO_I$	$ s_{(I)}^2 (X_{(I)}^T X_{(I)})^{-1} / s^2 (X^T X)^{-1} $
5. Small $COVRATIO_I$	$ s_{(I)}^2 (X_{(I)}^T X_{(I)})^{-1} / s^2 (X^T X)^{-1} $
6. C_I	$(\hat{\beta} - \hat{\beta}_{(I)})^T X^T X (\hat{\beta} - \hat{\beta}_{(I)}) / ps^2$
7. WK_I	$(\hat{\beta} - \hat{\beta}_{(I)})^T X^T X (\hat{\beta} - \hat{\beta}_{(I)}) / ps_{(I)}^2$
8. W_I	$(\hat{\beta} - \hat{\beta}_{(I)})^T X_{(I)}^T X_{(I)} (\hat{\beta} - \hat{\beta}_{(I)}) / ps_{(I)}^2$

* - $Z = (X \ y)$

TABLE 2: The smallest ranking of influential set of size m that is required for searching the most influential set of size $m+1$

Data set	ME ^b	m		Data set	ME	m	
		2	3			2	3
AT4 ^a	1	2	-	CH129	5	-	2
17×2	6	2	11	30×5	7	3	5
	8	2	5				
AT49	3	-	2	CH208	6	-	2
28×4	8	-	2	49×7	7	-	2
AT53	1	5	2	CW4 ^c	-	-	-
32×5	3	3	-	24×11			
	4	2	2				
	6	30	4	CW22	-	-	-
	7	-	2	21×2			
	8	2	-				
BKH41	-	-	-	CW47	5	2	2
50×5				22×6	6	18	-
					7	2	-
					8	2	-
CH61	3	3	5				
20×7	6	-	6	CW119	8	3	3
	7	-	25	19×4			
CH82	-	-	-				
18×3				CW149 ^d	1	4	2
				31×10	2	2	68
					3	5	2
					5	2	56
CH89	3	2	-		6	2	32
31×6					7	2	35
					8	2	85

a - AT4: Atkinson (1985) p. 4, 17×2 = number of observations $n \times$ number of predictor variables p , BKH: Belsley, Kuh and Welsch (1980), CH: Chatterjee and Hadi (1988), CW: Cook and Weisberg (1982).

b - ME - Measure. Label is as given in table 1.

c,d - near or exact multicollinearity after the omission of observations 3 and 20 for CW4, and observations 5, 6, 9 and 26 for CW149 respectively.

TABLE 3: Four most influential measure values and sets with size m for pollution data

ME	m							
	1		2		3		4	
1	(21)	0.219	(21,67) ^a	0.363 ^b	(21,67;62)	0.468	(21,67;62;59)	0.553
	(67)	0.184	(21,62)	0.347	(21,67;59)	0.458	(21,67;62;4)	0.546
	(62)	0.156	(21,59)	0.336	(21,67;114)	0.454	(21,67;62;114)	0.544
	(4)	0.125	(21,4)	0.317	(21,62;59)	0.451	(21,67;59;114)	0.537
2	(21)	0.169	(21,86)	0.253	(21,86;85)	0.331	(21,86;85;97)	0.371
	(86)	0.094	(21,85)	0.239	(21,86;97)	0.289	(21,86;85;20)	0.364
	(85)	0.085	(21,97)	0.205	(21,86;20)	0.285	(21,86;85;57)	0.360
	(97)	0.036	(21,20)	0.201	(21,86;57)	0.283	(21,86;85;26)	0.359
3	(21)	0.357	(21,67)	0.497	(21,67;85)	0.588	(21,67;85;62)	0.657
	(67)	0.215	(21,85)	0.473	(21,67;62)	0.580	(21,67;85;114)	0.656
	(85)	0.176	(21,62)	0.463	(21,67;114)	0.576	(21,67;85;86)	0.652
	(62)	0.157	(21,59)	0.454	(21,67;59)	0.574	(21,67;85;59)	0.651
4	(62)	1.24	(62;59)	1.47	(62;59;114)	1.72	(62;59;114;112)	2.00
	(59)	1.18	(62;114)	1.45	(62;59;112)	1.71	(62;59;114;100)	1.99
	(114)	1.17	(62;112)	1.44	(62;59;100)	1.70	(62;59;112;100)	1.99
	(112)	1.14	(62;100)	1.42	(62;114;112)	1.68	(62;59;112;4)	1.97
5	(21)	0.507	(21,86)	0.307	(21,86;85)	0.195	(21,86;85;97)	0.147
	(86)	0.627	(21,85)	0.359	(21,86;97)	0.245	(21,86;85;20)	0.155
	(85)	0.706	(86;85)	0.402	(21,86;20)	0.254	(21,86;85;19)	0.165
	(97)	0.849	(21,97)	0.416	(21,86;26)	0.265	(21,86;85;26)	0.166
6	(21)	1.110	(21,57)	1.66	(21,57;59)	2.04	(21,57;59;100)	2.54
	(85)	0.210	(21,19)	1.39	(21,57;100)	2.02	(21,57;59;19)	2.52
	(67)	0.195	(21,20)	1.37	(21,57;19)	2.02	(21,57;59;20)	2.46
	(86)	0.060	(21,105)	1.34	(21,57;20)	1.97	(21,57;59;105)	2.44
7	(21)	1.330	(21,57)	2.08	(21,57;19)	2.60	(21,57;19;20)	3.28
	(85)	0.230	(21,19)	1.73	(21,57;20)	2.58	(21,57;19;59)	3.26
	(67)	0.202	(21,20)	1.72	(21,57;59)	2.52	(21,57;59;20)	3.20
	(86)	0.067	(21,105)	1.64	(21,57;100)	2.52	(21,57;59;100)	3.16
8	(21)	1.04	(21,57)	1.54	(21,57;19)	1.91	(21,57;19;20)	2.36
	(85)	0.208	(21,19)	1.33	(21,57;20)	1.89	(21,57;19;105)	2.24
	(67)	0.164	(21,20)	1.33	(21,57;105)	1.81	(21,57;20;105)	2.24
	(86)	0.065	(21,105)	1.25	(21,57;102)	1.72	(21,57;19;100)	2.10

ME - Measure, the measures are exactly the same as those in table 1.

a - observations 21 and 67 form the most influential set for the measure.

b - measure values.